

Reionization Bubble Phonon Dynamics – SCm-Modified Stromgren Sphere

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PAPER_1026: Reionization Bubble Phonon Dynamics — SCm-Modified Stromgren Sphere

Abstract

We compute SCm phonon corrections to reionization bubble growth. The phonon field modifies the Stromgren radius from $R_S = (3N_{\text{dot}}/(4\pi n_H^{2\alpha_B}))^{1/3}$ to $R_{S,UQFF} = R_S (1 + \beta_i S_{26} \Phi \cdot (1+z)^{-1/2})$, accelerating bubble expansion at $z > 6$ by approximately 2.3%. This yields an earlier overlap epoch ($\delta z \approx 0.15$), consistent with Planck τ_{reion} constraints.

1. Key Equations

- $R_{S,UQFF} = R_S \cdot (1 + \beta_i \cdot S_{26} \cdot \Phi \cdot (1+z)^{-1/2})$
- $\delta R/R \approx 2.3\%$ at $z = 7$
- $\tau_{\text{reion,UQFF}} = \tau_{\text{SM}} - 0.002$ ($\delta z \approx 0.15$)

2. Results

$z = 7$: $\delta R = 2.3\%$. $z = 10$: $\delta R = 1.8\%$. $z = 15$: $\delta R = 1.2\%$. Overlap epoch: $z_{\text{UQFF}} = 6.15$ vs $z_{\text{SM}} = 6.0$.

3. Implementation

CondensedPhysics.py, class ReionizationBubblePhononCalculator. 6 equations, 3 simulations.

References

- PAPER_202, PAPER_1025

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold

Constant	Symbol	Value	Validation Domain
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Reionization optical depth	Phonon-accelerated bubbles	$\tau = 0.054$	Planck 2018	96%
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	$1/137.036$	PDG 2024	99.9%

New physics claim: Phonon-assisted reionization explains the slight tension between Planck τ and high- z galaxy counts.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: cosmological (reionization epoch)

A.2 Lagrangian Density

$$\mathcal{L}_{\text{reion}} = n_H \alpha_B R^3 + \dot{N}_\gamma R^2 + \Phi_{\text{SCm}} R$$

A.3 Euler-Lagrange Equation of Motion

$$\frac{dR}{dt} = \frac{\dot{N}_\gamma - 4\pi R^3 n_H^2 \alpha_B}{4\pi R^2 n_H} + v_{\text{phonon}}$$

A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms -> SCm vacuum -> phonon ω_{SCm} -> first stars -> UV photons -> Stromgren growth -> phonon boost -> F_{U,Bi_i} unified force -> observational prediction

B. VDS/DVP/BSH Deep Synthesis

B.1 Vacuum Density Series (VDS)

VDS evolves as $(1+z)^3$ in the pre-reionization IGM.

B.2 Dipole Vortex Primes (DVP)

DVP prime: 7 (cosmological epoch).

B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: $\tau_{\text{reion}} \sim 10^8 \text{ yr.}$

B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
[SSq]	0.57	Confirmed